

Prabhash Thakur

Senior Lead Engineer

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SUMMARY

Data engineer with 8+ years on Snowflake and dbt, currently leading a team of 8 at KIPI.AI building a Data Mesh and Gen AI data products with Snowflake Cortex. Recent work covers RAG applications over unstructured data (PDFs, DOCX, audio transcripts) using Cortex Search, self-service analytics through Cortex Analyst, and LLM-driven enrichment pipelines with Cortex functions. Comfortable picking the right modelling approach for the problem (Data Vault, star schema, medallion), locking down access with RBAC and masking policies, and keeping Snowflake compute costs under control. Also handle architecture decisions and work directly with client stakeholders on scope and delivery. SnowPro Advanced Architect, SnowPro Advanced Engineer, SnowPro Core, and dbt Developer certified.

SKILLS

Data Platform & Engineering: Snowflake, dbt, SQL, Python, Azure Data Factory, Databricks, Azure Data Lake, AWS S3, Airflow, Jinja, ETL/ELT Pipeline Design, Data Warehouse Architecture

Snowflake AI & Gen AI: Cortex COMPLETE, CLASSIFY_TEXT, EXTRACT_ANSWER, SENTIMENT, SUMMARIZE, TRANSLATE, AI_PARSE_DOCUMENT, Cortex Search (RAG), Cortex Analyst, Cortex Fine-Tuning, Document AI, Streamlit in Snowflake, Unstructured Data (PDF, DOCX, Audio)

Architecture & Governance: Data Mesh, Medallion Architecture, Data Vault Modeling, Dimensional Modeling, Star Schema, Data Quality Frameworks, dbt-expectations, Row-Level Security, RBAC, Dynamic Data Masking, Data Lineage, CI/CD for dbt, Azure DevOps, Git

Reporting & Automation: Power BI, DAX, M Query, Power Automate, PowerApps, Azure Logic Apps, Azure Functions, SAP BW, SAP HANA

Leadership & Delivery: Team Leadership (8+ engineers), Mentorship, Stakeholder Management, Sprint Planning, Delivery Planning, Cross-Functional Governance

EXPERIENCE

Senior Lead Engineer

July 2025 - Present

KIPI.AI

- Lead a team of 8 building a **Data Mesh** on Snowflake for KIPI's largest client, with data products across 5+ domains used by around 200 people.
- Built a **RAG-based Q&A chatbot**. Used **AI_PARSE_DOCUMENT** to pull text out of PDFs, DOCX files and audio transcripts, chunked and indexed it into Cortex Search, and used Cortex COMPLETE to generate the answers. Shipped as a Streamlit app in Snowflake. Query resolution went from hours to seconds and new joiners get up to speed faster.
- Automated classification of 50K+ customer feedback records a month coming from free-text survey responses, support emails and call recordings. Added **Cortex LLM functions** (CLASSIFY_TEXT, SENTIMENT, EXTRACT_ANSWER) into the enrichment pipeline, which freed up 3 people and turned a weeks-long backlog into same-day output.
- Set up **Cortex Analyst** so business users could ask questions in plain English instead of raising SQL requests. Ad-hoc report requests dropped by around 40%.
- Used **Document AI** to pull amounts, dates, vendor names and line items out of vendor invoices and contracts, feeding straight into the finance data product. Cut roughly 20 hours a week of manual data entry and sped up accounts payable processing.
- Fine-tuned a base LLM with **Cortex Fine-Tuning** on client-specific labels, which improved classification accuracy by around 15% over the out-of-the-box model.

- Cut compute cost on the main ingestion pipeline by 60%. Moved it to incremental loads with micro-batching and rewrote the SCD2 logic so it runs on an XS warehouse instead of an L, on over 1TB a day.
- Wrote macros that purge stale snapshots automatically, cutting storage and processing overhead by around 40%.
- Set up coding standards, PR review process, branching strategy and naming conventions. None of this was consistent before I joined.
- Built a PII scanner that checks every data product for missing Row-Level Security or masking policies and alerts the domain owner when it finds a gap.
- Built the data quality layer with **dbt-expectations** and am putting together a wider governance framework so quality holds up as new domains come on.
- Run weekly sessions with the team on code walkthroughs, architecture and practices.
- Main point of contact for 3 client stakeholder groups. Handle sprint planning, backlog grooming and delivery timelines.

Senior Technical Lead

June 2023 - July 2025

Beyond Key

- Led 8+ engineers on Snowflake migrations and new builds for enterprise clients.
- Brought **dbt** into client projects from nothing. Set up the project structure, wrote the standards documentation and built CI/CD through Azure DevOps for clients who had never used dbt.
- Designed ETL/ELT pipelines on Snowflake using **medallion architecture** (bronze/silver/gold) and star schema modelling, turning raw data into datasets analysts could use directly.
- Built **ADF pipelines** with dynamic schema detection that created Snowflake stages and tables on their own, so nobody had to hand-write DDL for each new source.
- Wrote JavaScript stored procedures in Snowflake to set up stages and Snowpipes from config table entries.
- Set up **Secure Data Sharing** for cross-organization distribution, plus RBAC, dynamic masking and tag-based masking on sensitive columns.
- Wrote reusable macros for error logging on failed dbt models and for handling insert and update timestamps consistently across projects.
- Wrote custom dbt tests for referential integrity and source-to-target row count checks.
- Ran internal Snowflake and dbt training for 20+ engineers and led the Snowflake practice POC for Beyond Key.
- Rebuilt an existing ADF extraction pipeline into a modular design so the team could add new sources without touching core logic.
- Built export tasks to push data out of Snowflake to downstream consumers.

Senior Data Engineer

June 2021 - June 2023

Visual BI Solutions (an Atos Company)

- Built Snowflake ETL pipelines with dbt across several client projects, using **Data Vault** or star schema depending on what each client needed for reporting.
- Wrote reusable macros for **SCD Type 1 and Type 2** across mixed column types, which removed per-table models and cut repeated code by around 60%.
- Tracked down a null primary key problem in left joins caused by late-arriving dimension records. Prototyped the fix, validated it with the client and put it into production.
- Worked with every dbt materialization (table, view, incremental, snapshot, ephemeral) and handled dbt version upgrades, testing impact before each one.
- Set up data quality testing with **dbt-expectations**, getting 95%+ coverage on critical models with checks for region filtering, row count matching and nulls.
- Optimized complex SQL using CTEs, window functions and recursive queries. Overrode global dbt macros so full-refresh could switch between truncate-reload and drop-recreate at runtime.
- Designed a recovery process to rebuild historical SCD data from raw sources when snapshots got corrupted, bringing recovery time down from days to under 4 hours.
- Built an **RBAC automation tool** with PowerApps, Power Automate and Azure Functions, replacing manual SQL provisioning with a self-service form and cutting provisioning time by around 80%.

- Built Power BI dashboards on warehouse credit usage, which let leadership spot 25%+ wasted compute and resize warehouses.
- Owned the Git branching strategy in Azure DevOps for a dbt project of around 200 models.
- Streamed MSFT stock prices through ADF every minute and stored the high for each 5-minute window in Azure SQL, as part of internal COE work.

Software Engineer

March 2018 - May 2021

Tech Mahindra Ltd.

- Built out the Snowflake data warehouse for the project and wrote the dbt and SQL transformation logic behind 50+ downstream reports.
- Ran 15+ data pipelines on **SAP BW** (ADSOs, DSOs, Composite Providers) and automated scheduling with Process Chains, cutting manual data prep by around 30%.
- Moved legacy reports to **SAP HANA** calculation views, which ran around 40% faster than the BW versions.
- Built Power BI dashboards with DAX and M Query for 3 business units, replacing weekly Excel reporting.
- Owned deployments across dev, QA and production. Kept 99%+ on-time delivery by watching Process Chains and fixing failures before they escalated.
- Wrote data cleansing scripts to standardize 10+ source tables with inconsistent formats, which cut downstream transformation failures.
- Coordinated transport management across all 3 environments, working with other teams to hit release windows.

CERTIFICATIONS

- **SnowPro Advanced Data Architect** - Snowflake
- **dbt Developer Certification** - dbt Labs
- **SnowPro Advanced Data Engineer** - Snowflake
- **SnowPro Core Certification** - Snowflake
- **Microsoft Certified Data Analyst (DA-100)** - Microsoft
- **Azure Data Fundamentals (DP-900)** - Microsoft

EDUCATION

B. Tech – Kalyani Government Engineering College, Kalyani, West Bengal | 2013-2017 | 7.28 DGPA

Higher Secondary – DAV Higher Secondary School, Asansol, West Bengal | 2011-2013 | 76%

Madhyamik (Class 10) – DAV Higher Secondary School, Asansol, West Bengal | 2011 | 76%

RECOGNITION

- **Star Performer Award (2x)** for client solution delivery
- **Pinnacle Award** for on-time delivery and positive client feedback
- **PAT ON BACK Award** for going above and beyond on project deadlines
- **Ninja Squad (2x)** for technical problem-solving
- **Eagle Squad** for consistent team delivery
- Multiple client commendations for hands-on leadership

LANGUAGES

English, Hindi